DORMATHON 2026 · TRACK 1 · PREDICT EARLY, DECIDE BETTER

Scam Guardian

A pause before the money moves.

On-device AI | Android + Web | Built for Malaysian seniors



Three Peas in a Pod

scam-guardian.vercel.app

github.com/51-Shenn/scam-guardian

01 — THE PROBLEM

Malaysians lost RM2.97 billion to online fraud in 2025.

0.2%

of 2025 losses was returned to victims that year (RM6.65 mil of RM2.97 bil).

42%

of online fraud cases were phone scams: 28,698 cases, Jan–Nov 2025.

27.7%

of losses fell on people aged 60+, who were only 8.3% of victims.

Sources: NST (Jul 2026); Malay Mail (Dec 2025; May 2024, early-2024 figures)

02 — THE GAP

In a coached scam, every security check passes.

A caller posing as police or the bank talks the victim through each transfer, live. To the system it looks legitimate:

- The right person
- On their own phone
- Entering the correct PIN

THE HARD PART

A slow user isn't a suspicious one.

Seniors hesitate and make mistakes naturally. We need to tell a confused user from a coerced one, before they press Confirm.

03 — THE SOLUTION

One transfer. Three readings. Three responses.

NORMAL

Carry on

Saved recipient, usual amount. The transfer goes straight through.

STRUGGLING

A clearer path

Back steps or help taps turn on Simple mode: one instruction at a time, larger controls.

COACHED

Take a moment

On a call, paying someone new: a 15-second pause asking if someone is directing this.

After the pause the choice is still theirs. Guardian never blocks someone's own money.

04 — HOW IT WORKS

Signals in, response out.

1 · SIGNALS

Situation
Is a phone call active?

Transaction
New recipient, amount against the usual

Behaviour
Back steps, help taps, a pasted account number

2 · DECIDE ON DEVICE

Rule layer
Deterministic and explainable: call + new recipient + paste or 3× usual amount

AI review
Frozen on-device model (logistic regression + XGBoost) scores the session every 5 s

3 · RESPOND

Normal
No interruption

Struggling
Simple mode

Coached
15 s pause, plus a family alert if high

05 — HIGH RISK

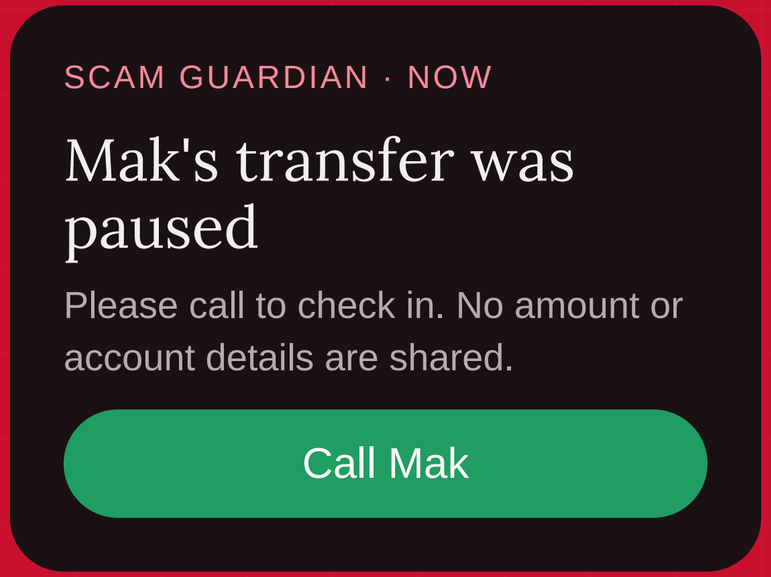
Bring in someone the scammer can't control.

On a call, paying a new recipient, with a pasted account number or more than 3× the usual amount:

01 A 15-second pause

02 Their trusted contact is asked to call

A call from family breaks the scammer's demand for secrecy.



From the demo: Aina's phone, the trusted contact

06 — PRIVACY BY DESIGN

Your decision. Your privacy.

NEVER COLLECTED



Call audio



Caller identity



Clipboard contents



Activity in other apps

On the device

Scoring runs on the phone, from a summary of the session.

One transfer at a time

From choosing a recipient to confirm or cancel. Nothing else.

Consent first

Guardian and the family alert each need their own opt-in.

07 — WHY IT'S DIFFERENT

Others measure risk. Guardian also decides the response.

Capability	Fraud vendors	Kill switch	Monzo	Scam Guardian
Tells confusion from coercion	No	No	No	Yes
Acts before money leaves	Yes	Opt-in only	Yes	Yes
Family alert from behaviour	No	No	Amount-based	Yes
Runs fully on the device	Server-side	n/a	n/a	Yes
Designed for Malaysian seniors	No	Partly	No	Yes

Fraud vendors: BioCatch and similar behavioural tools. Kill switch: Maybank Kill Switch and Money Lock, Touch 'n Go security features. Monzo: Trusted Contacts (UK).

Source: Scam Guardian research brief, competitor review

Try Scam Guardian

Explore how Guardian responds during a simulated bank transfer

62/62 tests · 2 on-device models · Android + web · money, calls and alerts simulated.



Scan to open the demo

scam-guardian-one.vercel.app



A pause before the money moves.

Help for the confused. Time for the coerced.

Try the demo · scam-guardian-one.vercel.app

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